

Common Trainee Misconceptions — RAG Beyond the Demo

1. "RAG eliminates hallucinations."

Why it happens: the word "grounded" implies safety, and demos usually show a clean success case. **Correction:** RAG grounds the *retrieval* step, not the *generation* step by default. Even with a perfectly relevant citation retrieved, the LLM can still paraphrase beyond it. This project's own scoping note in docs/EVALUATION.md states groundedness is currently measured at the retrieval level (did we retrieve the right section), not full answer-level groundedness — an honest limitation, not a solved problem. **How to teach it:** show Slide 13 and ask trainees to distinguish "a citation is attached" from "the answer only says what the citation says." Point to `calculate_payout` as the one place this project sidesteps the risk entirely — by never letting the LLM compute the number at all, rather than trusting grounding to prevent a wrong figure.

2. "More retrieved chunks always produce better answers."

Why it happens: intuitively, more context feels like more safety. **Correction:** raising `top_k` increases the chance of retrieving noisy or irrelevant chunks alongside relevant ones, can dilute the LLM's attention on the actually-relevant passage, and increases cost/latency. `RetrieveChunksUseCase` deliberately pulls a larger internal candidate pool (`top_k=20` per retriever) but returns a smaller final set (`top_k=5`) after fusion — more candidates for ranking, not more context for the LLM. **How to teach it:** Lab Task 3, Question 3 — have trainees reason through the tradeoff before revealing the answer.

3. "Embeddings understand meaning exactly like humans do."

Why it happens: "semantic similarity" sounds like comprehension. **Correction:** embeddings encode statistical patterns of co-occurrence and usage, not verified meaning. This project hit a direct, real consequence of this: cosine similarity between the query "What is the capital of France?" and insurance policy text sat in the same numeric range as similarity between two genuinely related policy passages — a documented property (anisotropy) of the embedding model, not an implementation

bug. **How to teach it:** Slide 15's real recorded scores. Ask trainees to guess the out-of-corpus score before revealing it — most guess much lower than the real value.

4. "If retrieval succeeds, the final answer must be correct."

Why it happens: retrieval success feels like the hard part is done. **Correction:** retrieval returning the right chunk is necessary but not sufficient — the LLM can still misread, under-weight, or contradict the retrieved evidence during generation. This is exactly why docs/EVALUATION.md separates retrieval hit-rate from answer-level groundedness as two different, not-yet-fully-unified metrics. **How to teach it:** Lab Task 4 — have trainees manually check whether the generated narrative in an adjudication run actually matches what its own cited evidence says, rather than assuming it does because a citation is present.

5. "Retrieved documents are trusted instructions the model should follow."

Why it happens: trainees new to LLM security assume all text in the prompt has equal authority. **Correction:** retrieved content must be treated as *data to summarize*, never as instructions to obey — this is the core defense against indirect prompt injection. This project's evaluation set includes an `adversarial_prompt_injection` case testing direct injection via the user's query, with the brief additionally requiring at least one **indirect** case (malicious text embedded inside an ingested document itself) as the more realistic and more dangerous variant. **How to teach it:** Stretch Challenge 3 — have trainees construct and actually test an indirect injection attempt, rather than only discussing the concept abstractly.